

ID Number 112514030

Name 李 志 强

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question. (3.5 points each question)

- 1) Which of the following expressions is in the sum-of-products form?
 A) $(AB)(CD)$ B) $(A + B)(C + D)$ C) $AB(CD)$ D) $AB + CD$
- 2) Which type of gate can be used to add two bits?
 A) NAND B) XNAND C) XOR D) NOR

1) D
 2) C

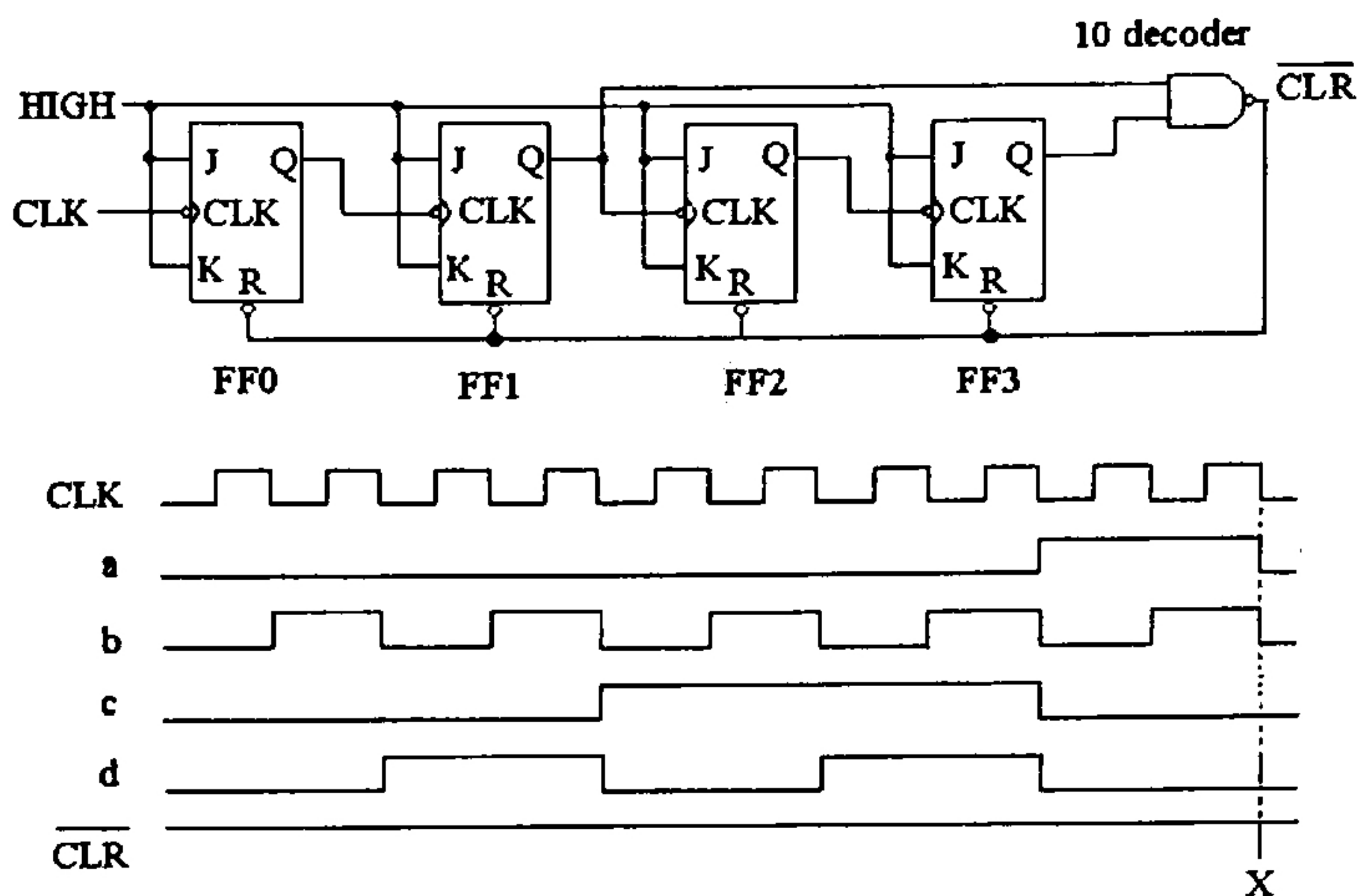


Figure 8-2

- 3) Which of the waveforms shown in Figure 8- represents the output of FF2?
 A) waveform a B) waveform b C) waveform c D) waveform d
- 4) Which of these truth tables represents the Exclusive-NOR gate?

3) C
 4) A

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

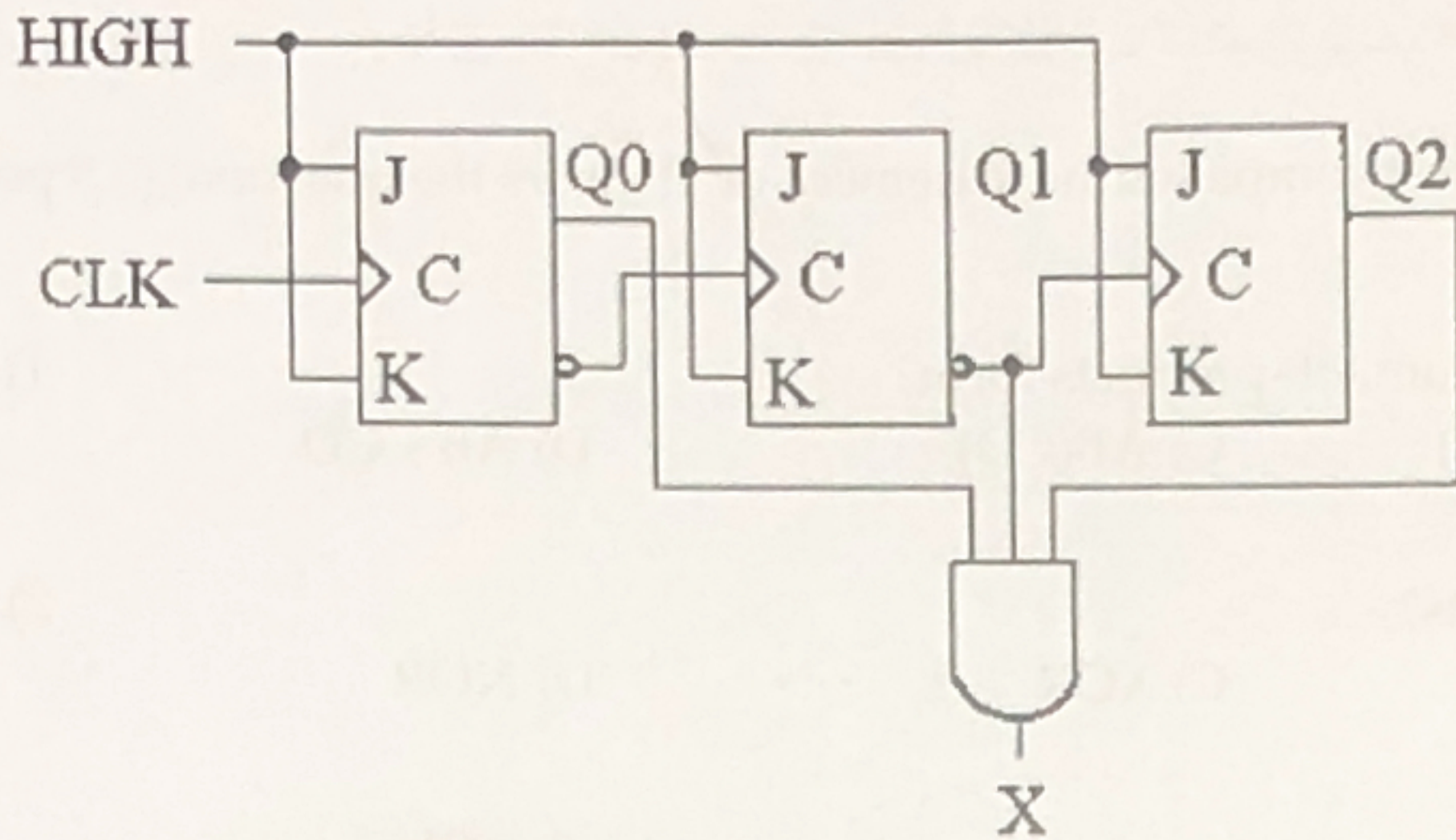
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

- (A) (A) (B) (B) (C) (C) D) (D)

5) What decimal value is required to produce an output at 'X' in this circuit?

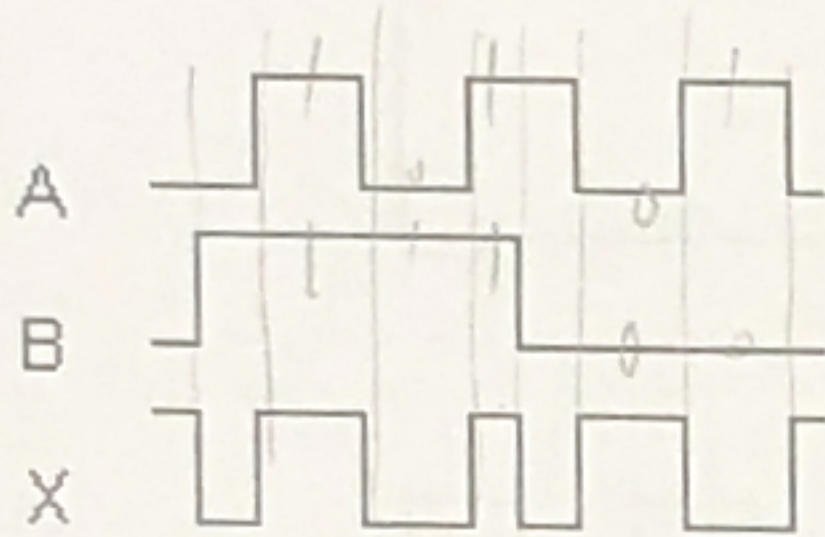
5) A



- A) 5 B) 2 C) 1 or 4 D) 1

6) This is the timing diagram for a 2-input _____ gate.

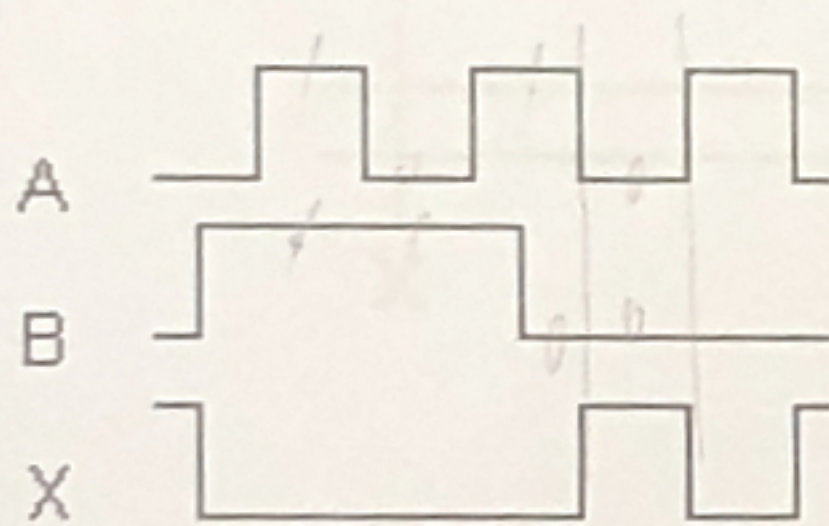
6) D



- A) AND B) Exclusive-OR
C) NAND D) Exclusive-NOR

7) This is the timing diagram for a 2-input _____ gate.

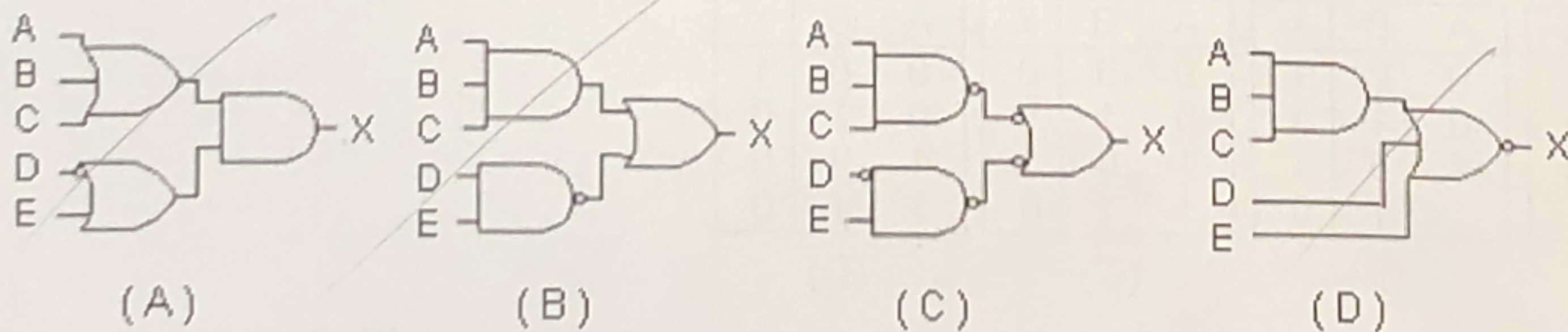
7) C



- A) OR B) Exclusive-NOR
C) NOR D) AND

8) Which of the figures is the correct NAND logic implementation of the expression, $X = ABC + \overline{DE}$?

8) C

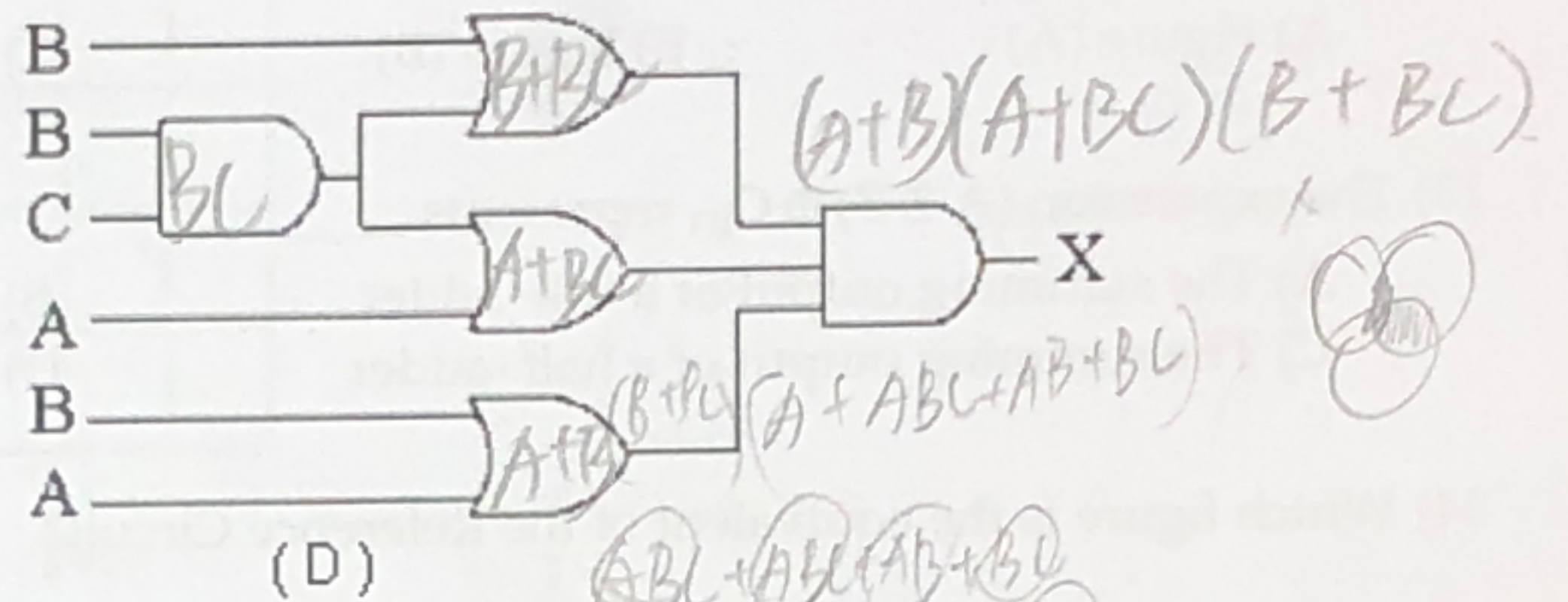
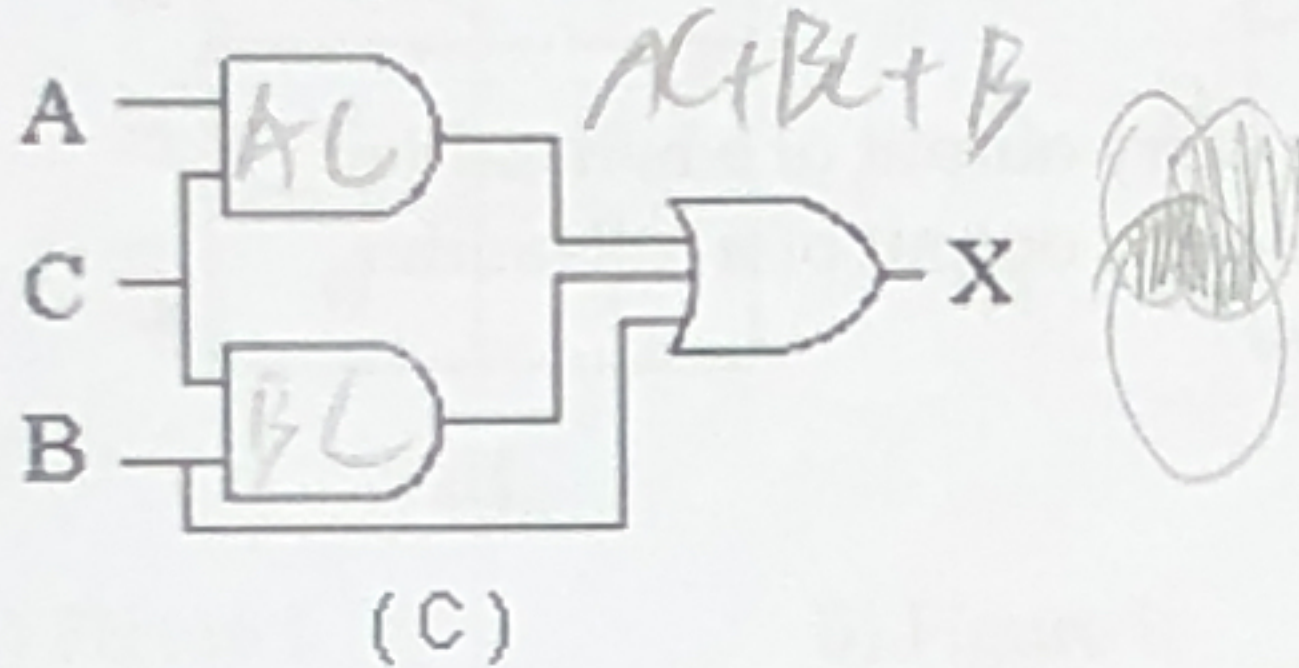
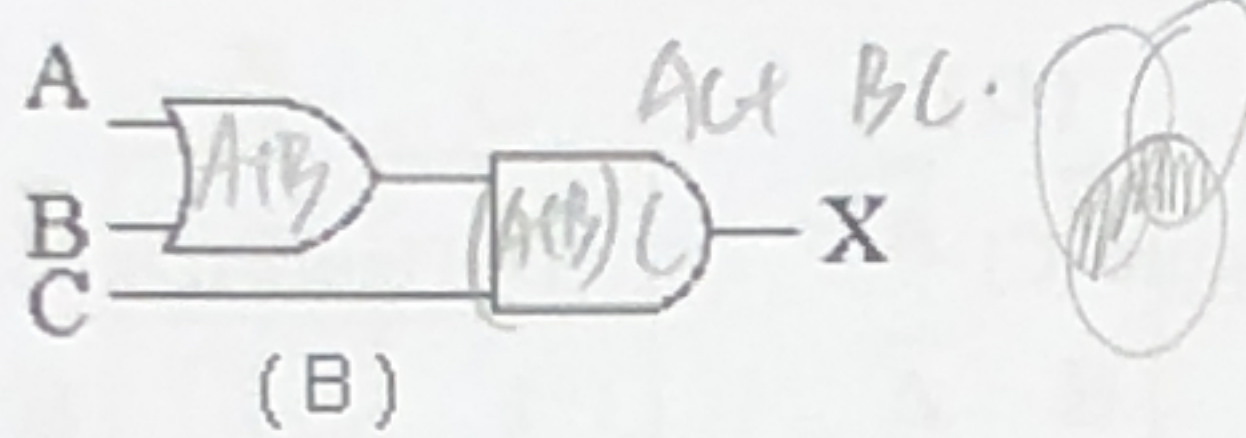
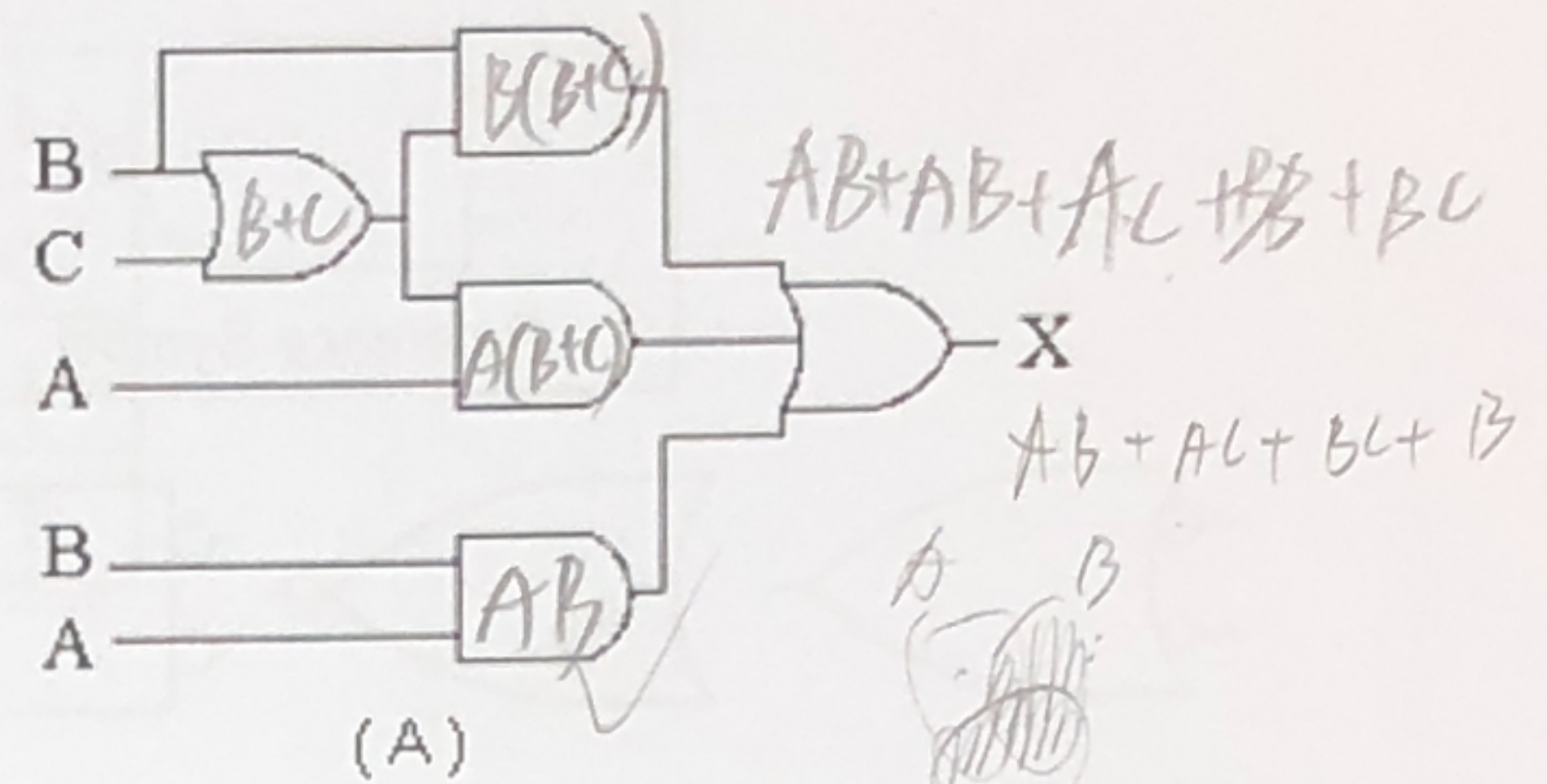
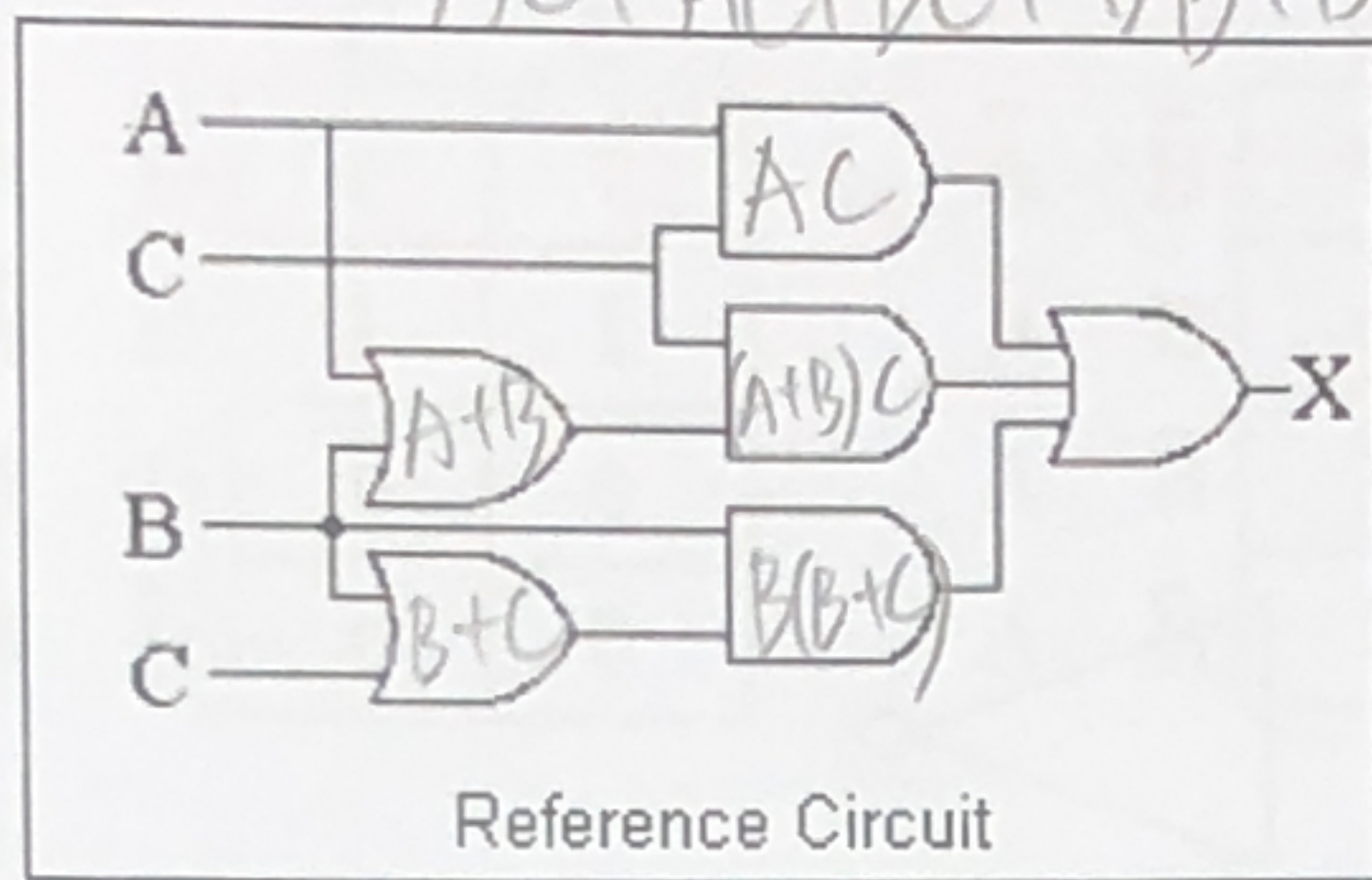


- A) Figure (A) B) Figure (B) C) Figure (C) D) Figure (D)

A
B
C

9) Which of the circuits below is equivalent to the Reference Circuit?

9) AC



A) Figure (A)

B) Figure (B)

C) Figure (C)

D) Figure (D)

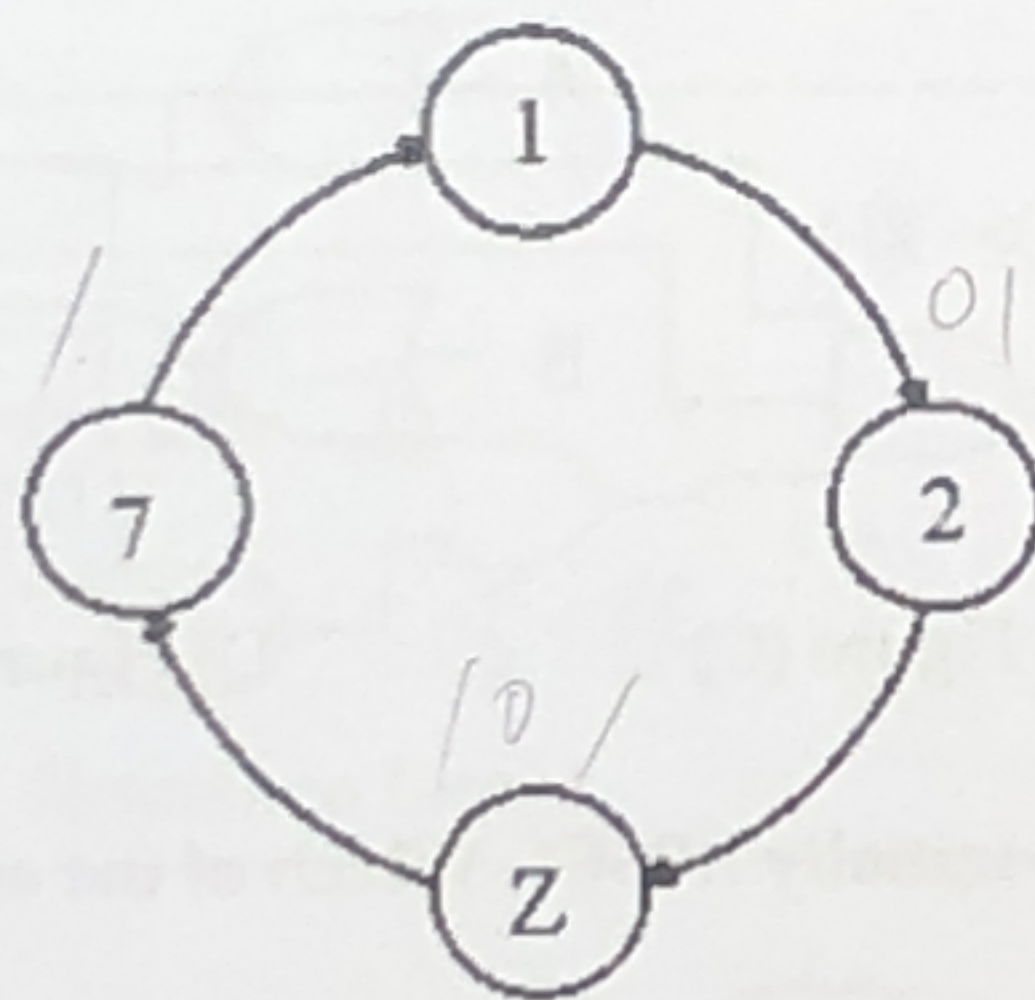


Figure 8-8

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	1	1	X
1	1	1	0	0	1

10) Refer to Figure 8-8. What value should 'X' represent in the state table?

10) 3

A) 0

B) 1

11) What value does 'Z' represent in Figure 8-8?

11) 3

A) 111

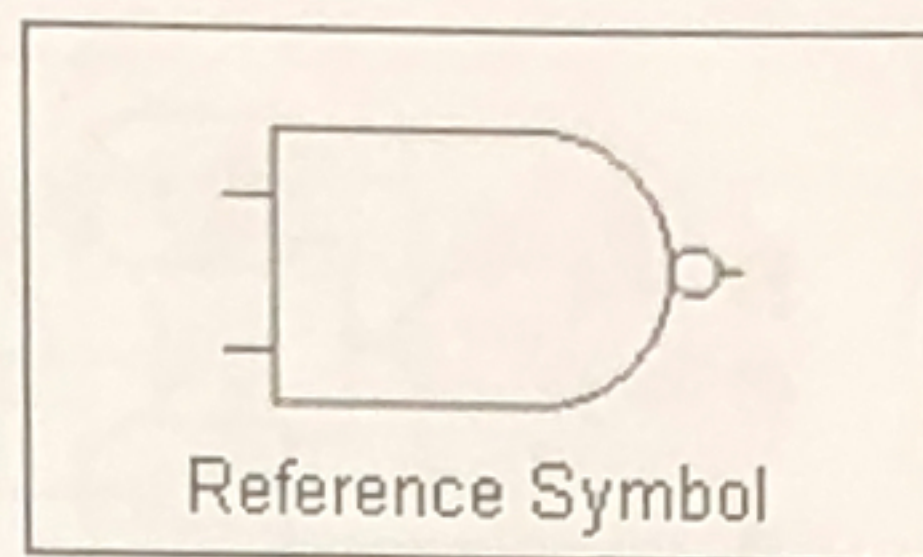
B) 5

C) 3

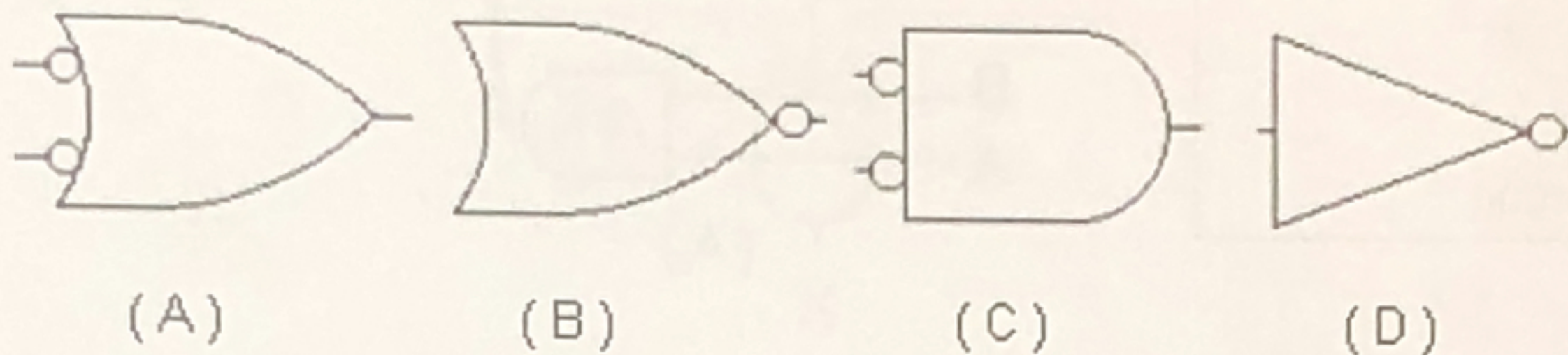
D) 4

12) Which of the figures is the DeMorgan equivalent of the Reference Circuit shown below?

12)



$$\overline{AB} = \overline{A} + \overline{B}$$



- A) Figure (A) B) Figure (B) C) Figure (C) D) Figure (D)

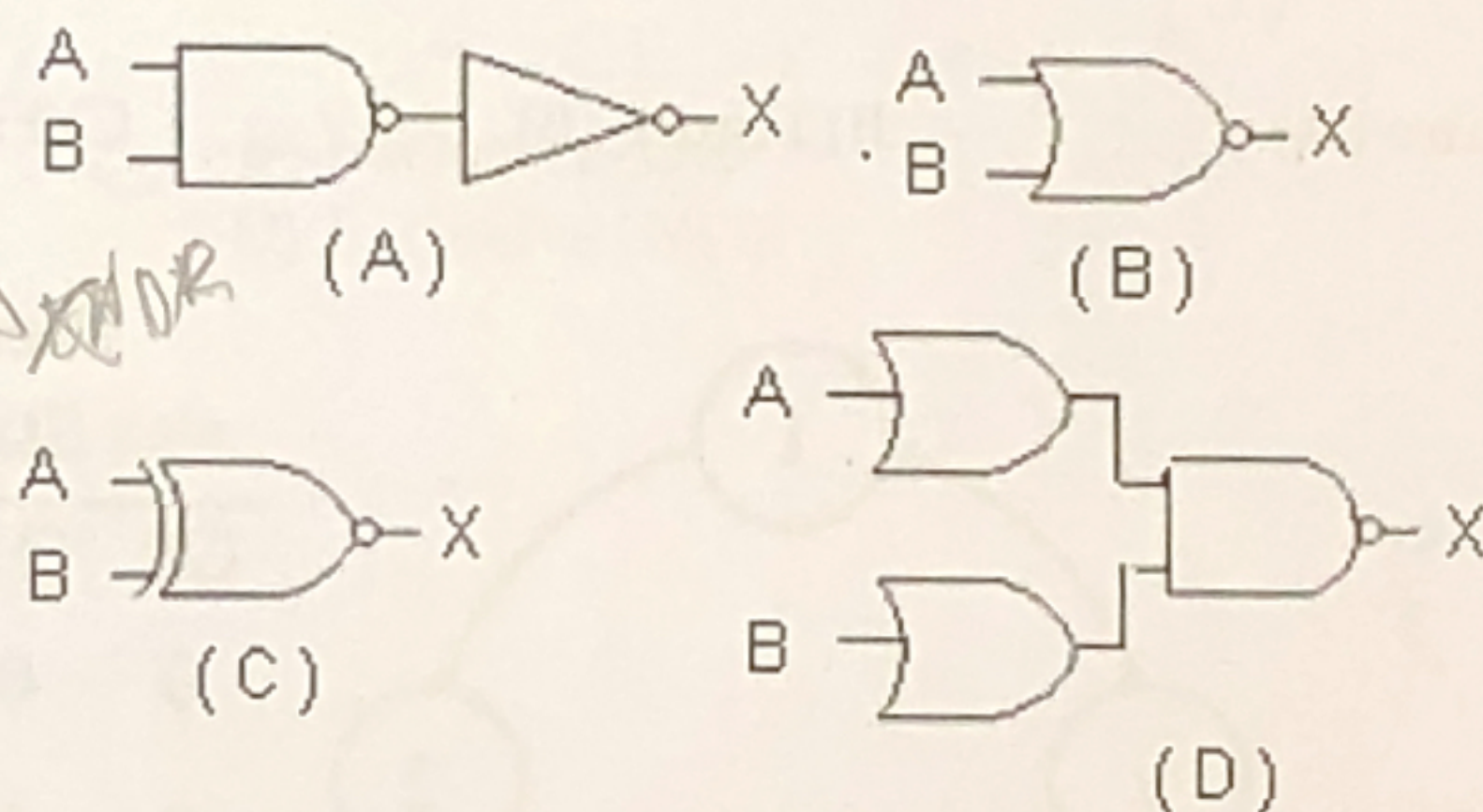
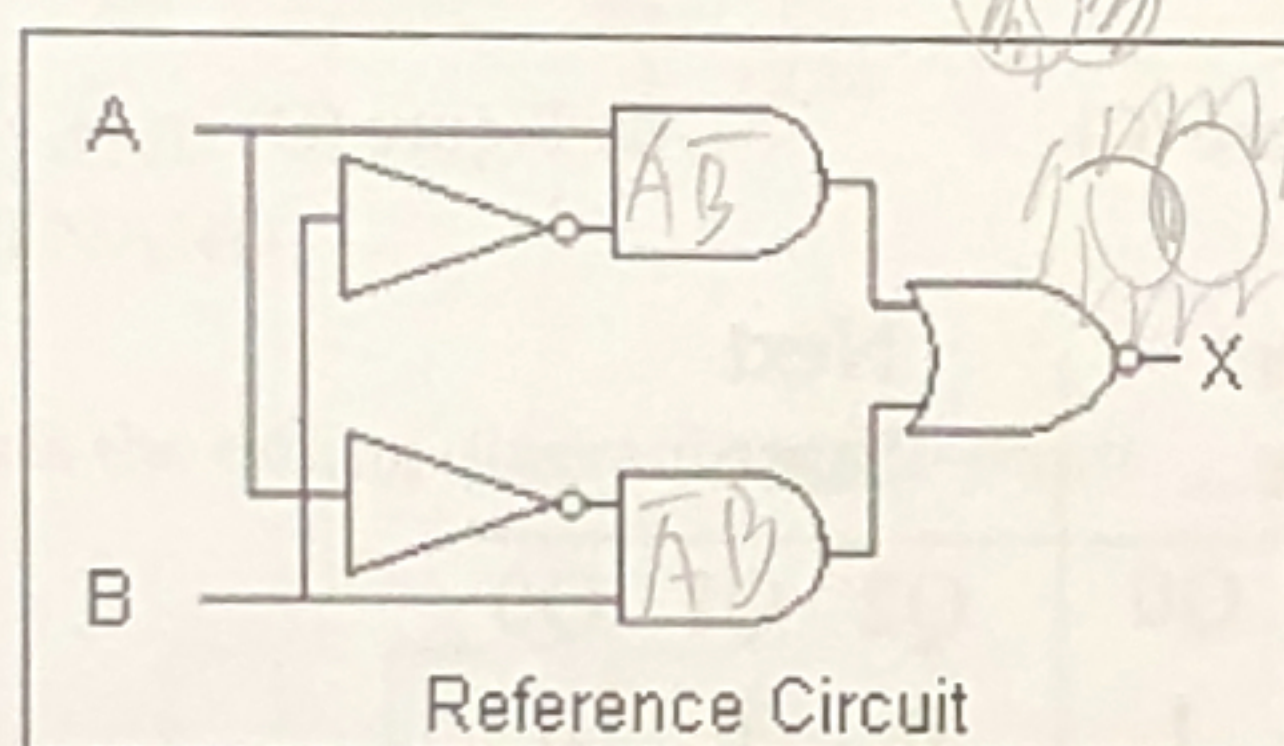
13) The expression $(A \oplus B) \oplus C_{in}$ represents _____.

13)

- A) The summing output of a full-adder B) The carry output of a half-adder
C) The summing output of a half-adder D) The carry output of a full-adder

14) Which figure is the equivalent of the Reference Circuit?

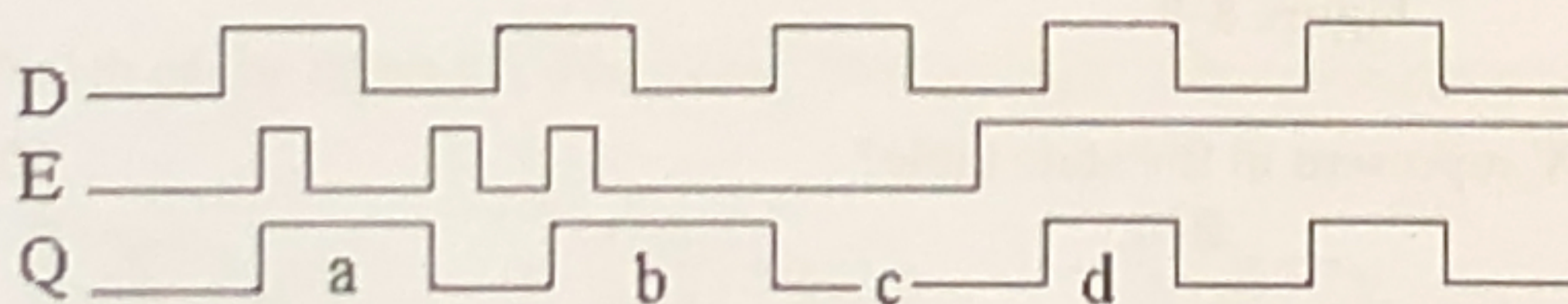
14)



- A) Figure (A) B) Figure (B) C) Figure (C) D) Figure (D)

15) These waveforms are applied to a gated D latch, which is initially RESET. Which of the areas identified on the Q waveform is incorrect?

15)



- A) Area a B) Area b C) Area c D) Area d

16) A J-K flip-flop is being used as a divide-by-2 circuit when _____.

16)

- A) the J and K inputs are tied to ground B) the reset is tied to the clock
C) the J and K inputs are tied to Vcc D) all the inputs are connected to the preset

17) Which of the Karnaugh maps below represents the expression, $X = AC + BC + B$?

17) C

$\bar{A}$	$\bar{B}$	$\bar{C}$	C
$\bar{A}$	$\bar{B}$	1	1
$\bar{A}$	B	1	1
A	B	0	0
A	$\bar{B}$	0	0

I.

$\bar{A}$	$\bar{B}$	$\bar{C}$	C
$\bar{A}$	$\bar{B}$	0	1
$\bar{A}$	B	0	0
A	B	1	1
A	$\bar{B}$	1	1

II.

$\bar{A}$	$\bar{B}$	$\bar{C}$	C
$\bar{A}$	$\bar{B}$	0	0
$\bar{A}$	B	1	1
A	B	1	1
A	$\bar{B}$	0	1

III.

$\bar{A}$	$\bar{B}$	$\bar{C}$	C
$\bar{A}$	$\bar{B}$	1	1
$\bar{A}$	B	0	1
A	B	0	1
A	$\bar{B}$	1	1

IV.

A) Figure I

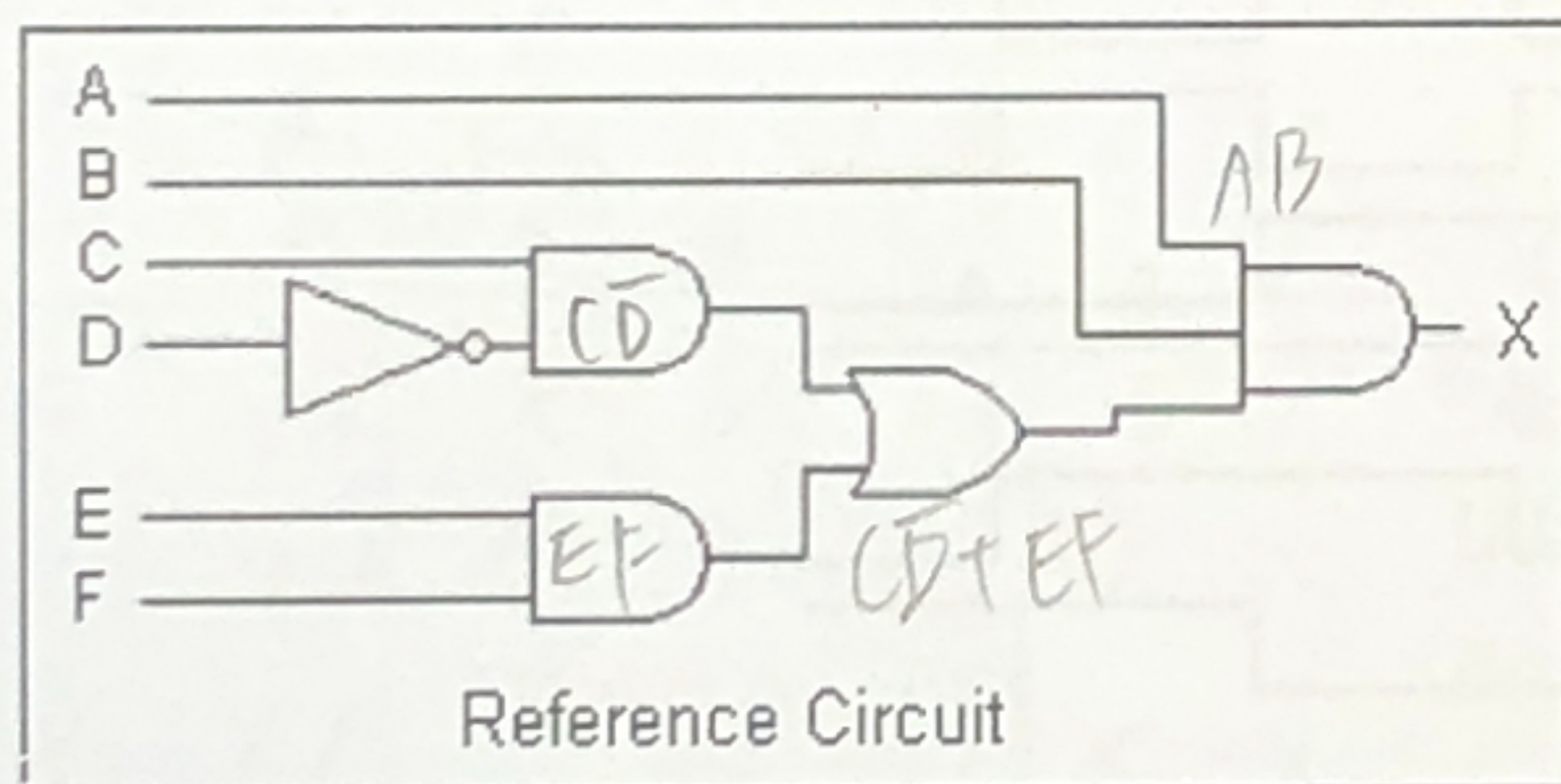
B) Figure II

C) Figure III

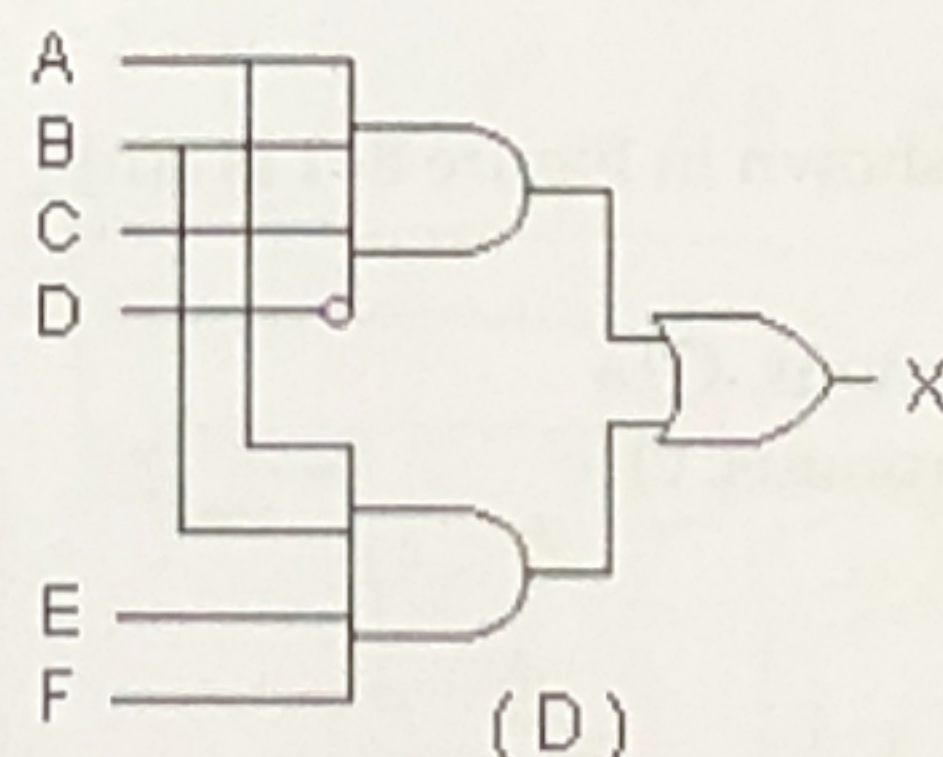
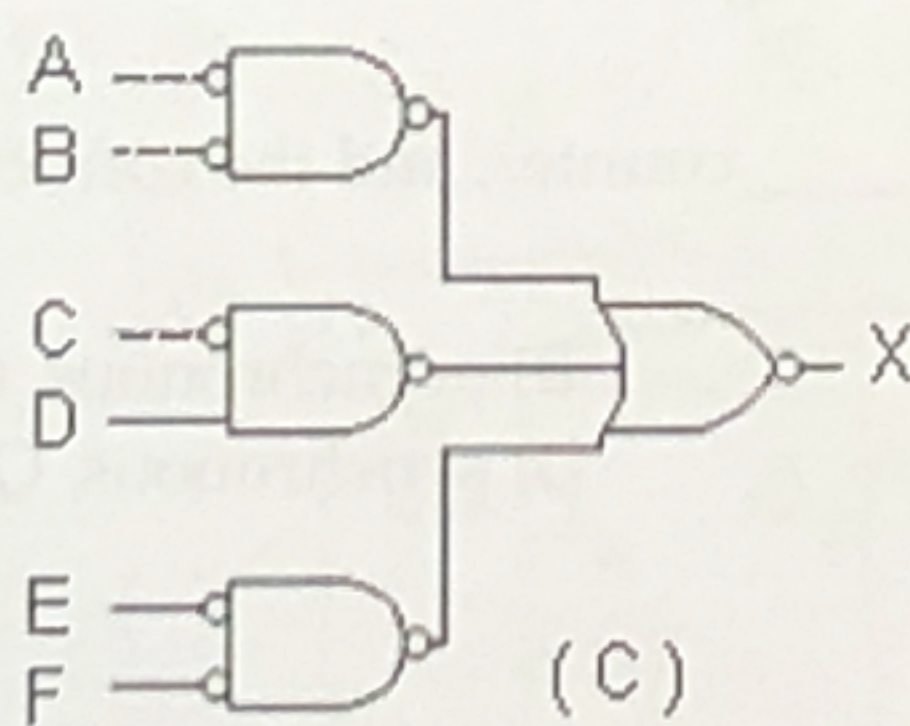
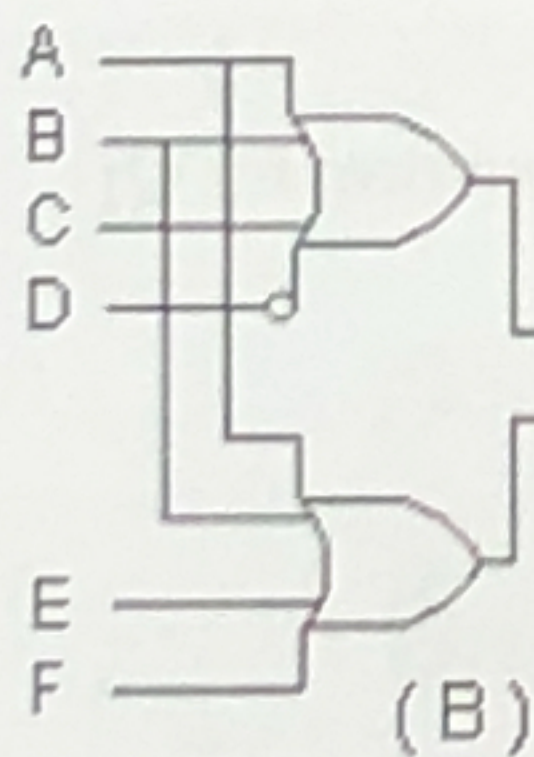
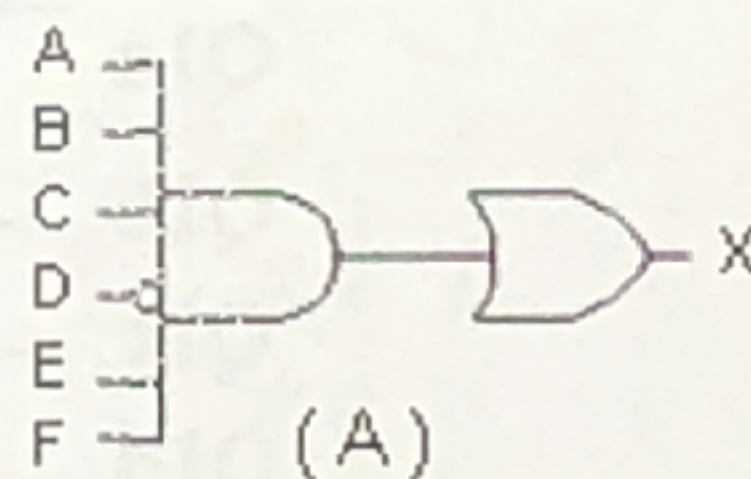
D) Figure IV

18) Which circuit is the sum-of-products equivalent of the Reference Circuit?

18) D



$$ABC\bar{D} + AB\bar{E}F$$



A) Figure(A)

B) Figure (B)

C) Figure (C)

D) Figure (D)

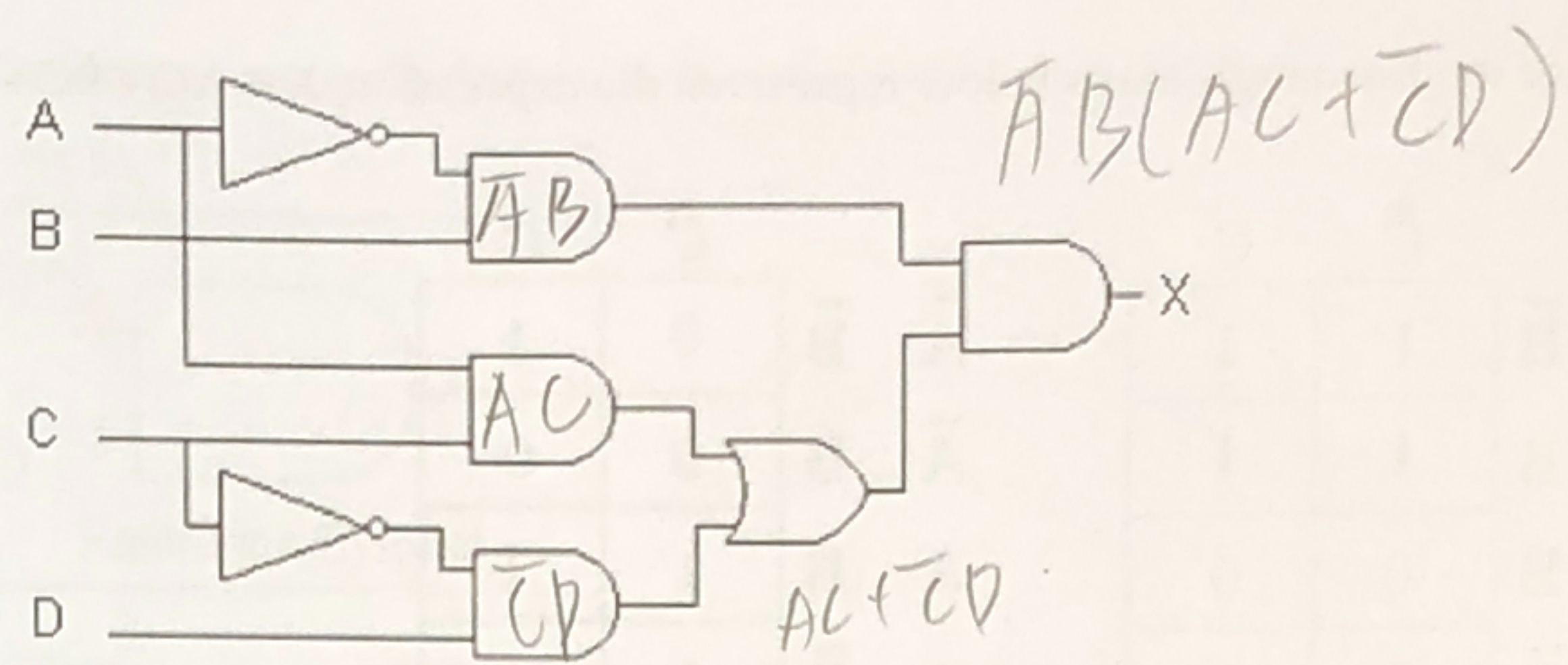


Figure 5-2

19) A correct logic expression for Figure 5-2 is _____.

- A) $X = (\overline{AB})(AC + \overline{CD})$
- B) $X = (\overline{AB})(A\overline{C}CD)$
- C) $X = ABC(\overline{C}BD)$
- D) $X = \overline{A}BC + A\overline{C}D$

19) A

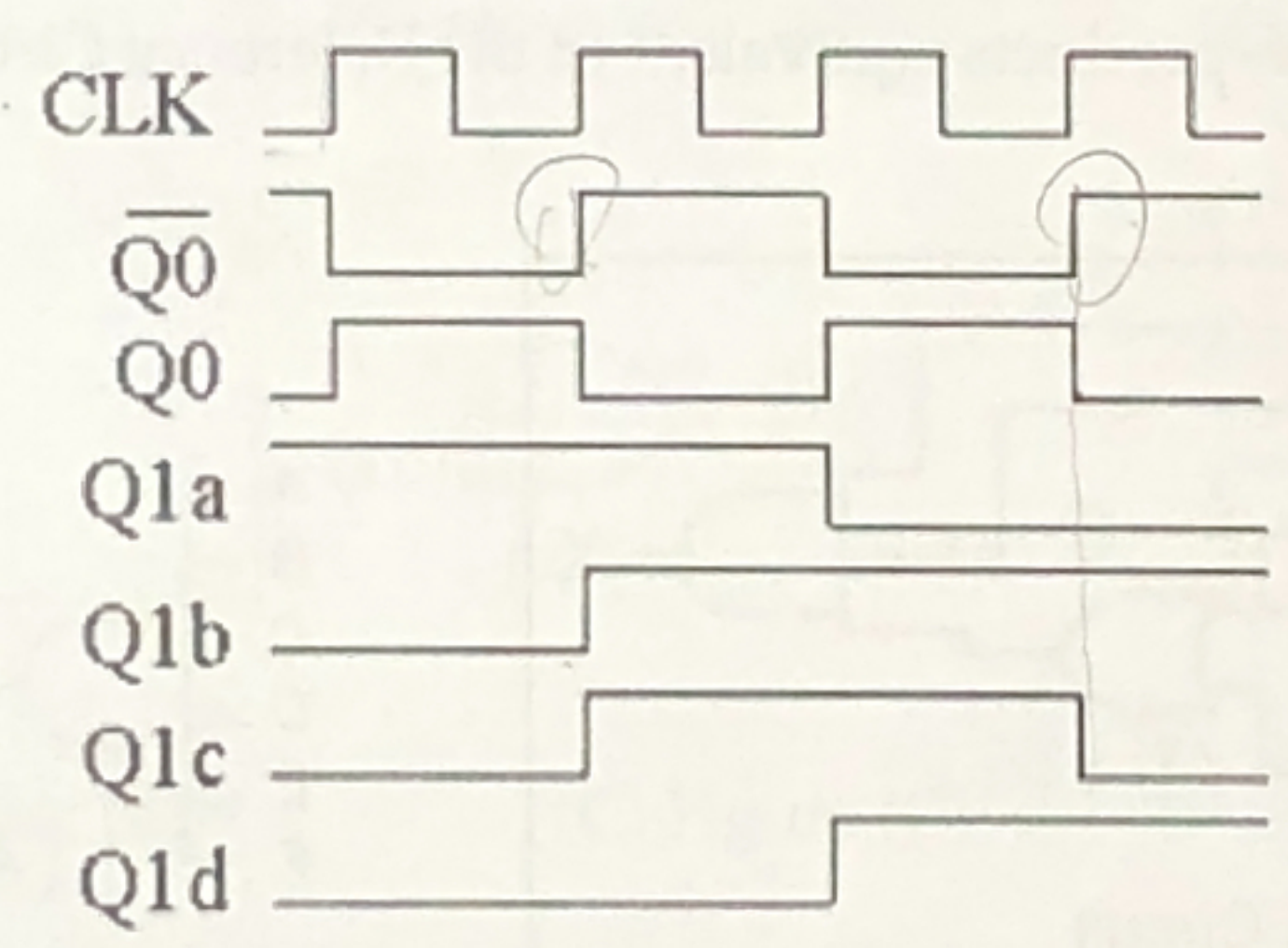
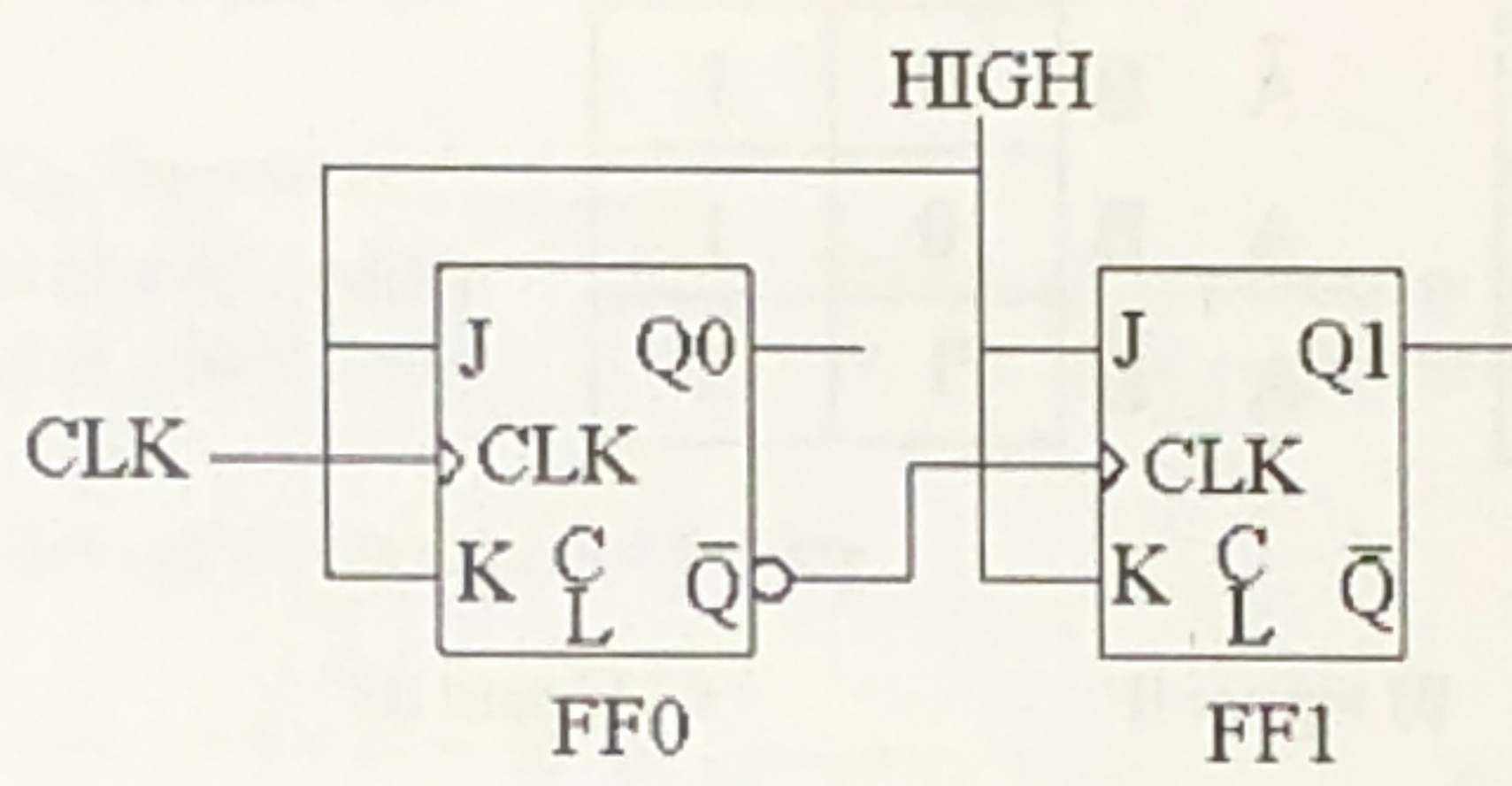


Figure 8-1

20) The counter shown in Figure 8-1 is a(n) _____ counter, and the correct output waveform for Q1 is _____.

- A) synchronous, Q1a
- B) asynchronous, Q1b
- C) asynchronous, Q1c
- D) synchronous, Q1d

20) C

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

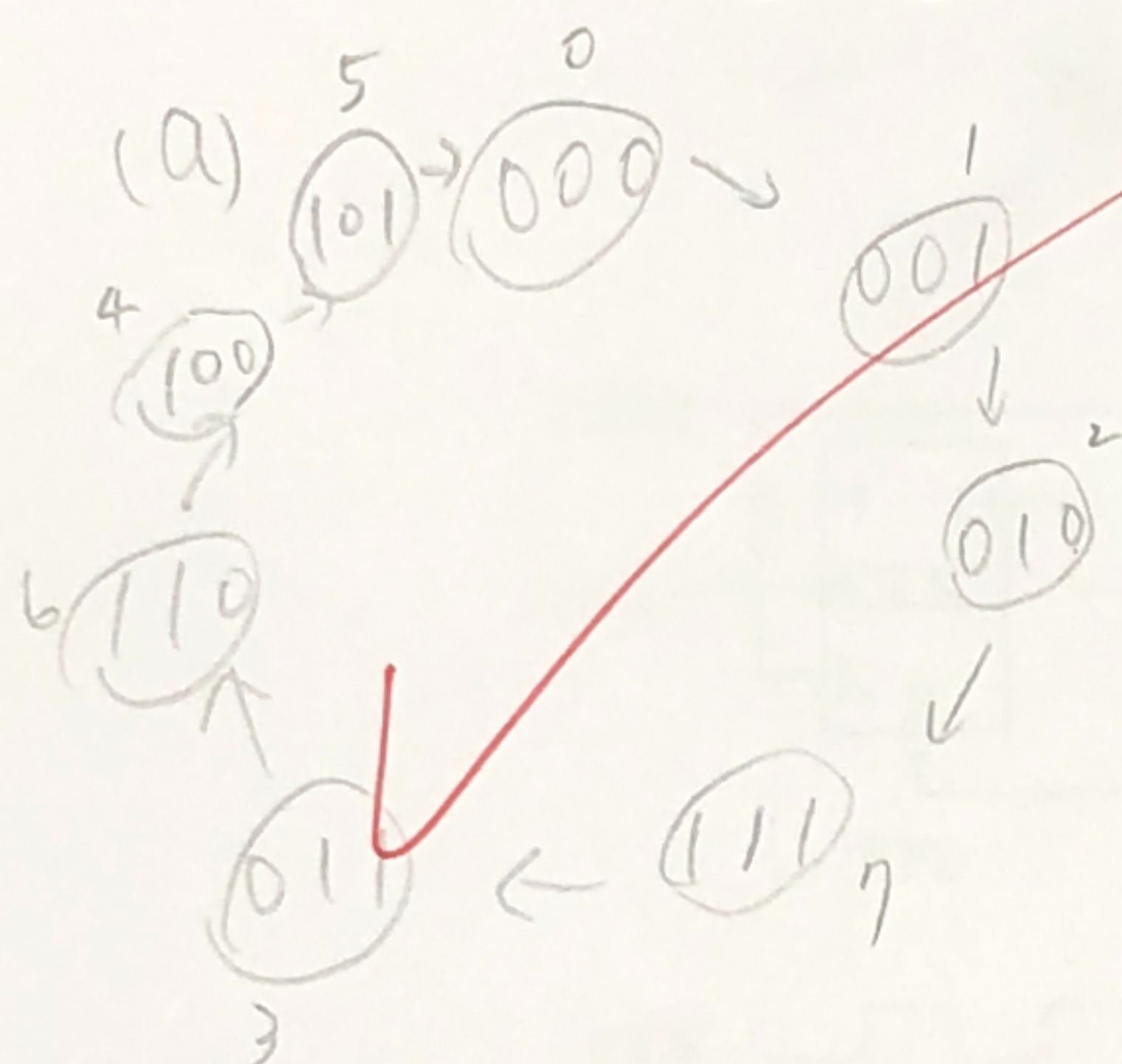
21) Design a synchronous counter to produce the following binary sequence. Use J-K flip-flops,

21) _____

0, 1, 2, 7, 3, 6, 4, 5, 0

- Draw the state diagram (6 points)
- List the next-state table (6 points)
- List the K-maps of all J's and K's (6 points)
- Write down the Boolean expressions of all J's and K's (6 points)
- Draw the logic diagram of your design (6 points)

1 1 → x 0
0 1 → x 1
0 1 → 1 x
0 0 → 0 x



(b) Next state table

Present state			Next state							
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	J_2	K_2	J_1	K_1	J_0
0	0	0	0	0	1	0	X	0	X	1
0	0	1	0	1	0	0	X	1	X	X
0	1	0	1	1	1	1	X	X	0	1
0	1	1	1	1	0	1	X	X	0	X
1	0	0	1	0	1	X	0	0	X	1
1	0	1	0	0	0	X	1	0	X	X
1	1	0	1	0	0	X	0	X	1	0
1	1	1	0	1	1	X	1	X	0	X

(c)

J_2				K_2				J_1				K_1				J_0				K_0			
$Q_2 \backslash Q_1$	0	1		$Q_2 \backslash Q_1$	0	1		$Q_2 \backslash Q_1$	0	1		$Q_2 \backslash Q_1$	0	1		$Q_2 \backslash Q_1$	0	1		$Q_2 \backslash Q_1$	0	1	
00	0	0		00	X	X		00	0	1		00	X	X		00	1	X		00	X	1	
01	1	1		01	X	X		01	X	X		01	0	0		01	1	X		01	X	1	
11	X	X		11	0	1		11	X	X		11	1	0		11	0	X		11	X	0	
10	X	X		10	0	1		10	0	0		10	X	X		10	1	X		10	X	1	

(d) Boolean expressions:

$$J_2 = Q_1$$

$$K_2 = Q_0$$

$$J_1 = \overline{Q_2} Q_0$$

$$K_1 = Q_2 \overline{Q_0}$$

$$J_0 = \overline{Q_2} + \overline{Q_1}$$

$$K_0 = \overline{Q_2} + \overline{Q_1}$$

